

ESILV DATA & AI · MASTER THESIS

Anomaly detection for data quality, **measured** inside the ETL pipeline

Rules, classical and neural detectors run through one pipeline and compared per anomaly type. Living progress deck, 2026-11-12.

Illustration, not data: a generated still life. One pad of 312 stands proud; the low light finds it.

17 DETECTORS 6 TYPES 152 DATASETS

Theo Bigand · github.com/theobigand/etl-anomaly-benchmark

THE QUESTION

Does machine learning in an ETL pipeline beat plain rules, and for **which anomalies**?

Rules are the industry default: auditable, cheap, and blind to what the author did not anticipate. Recent work puts ML detectors inside the flow, but on one algorithm and one dataset. This thesis builds the missing head-to-head benchmark and breaks the result down by anomaly type.

WHAT GENERIC BENCHMARKS LEAVE OUT

Pipeline. Algorithms are scored outside any pipeline; integration cost is not measured.

Rules. The declarative baseline that dominates production is absent.

Types. One aggregate score per dataset hides which family finds which defect.

Threshold. One flag threshold, not the precision-recall trade-off.

Benchmarks rank detectors. Few report their decisions

	REPORTS TIME ORDER?	REPORTS DECISIONS AT A THRESHOLD?
ADBench, 30 algorithms, 57 sets	○ not found (AUC only)	○ not found (AUC only)
Agyemang 2024, one simulated set	○ none	● precision, recall, F1
AnoShift, Kyoto 2006 to 2015	● IID, NEAR, FAR splits	○ not found (AUC only)
FedAD-Bench, federated deep	○ not found	● F1-optimal threshold
Sun et al. 2024, online thresholds	● shift in a stream	● error-rate guarantees
This thesis	● six fault types, controlled generator	● decisions at a threshold, cost

● found in the text I read ○ not found, which is not proof of absence. The scan is not systematic: this is a gap I did not find filled.

In the texts I read, ADBench and AnoShift report AUC only, and none of these works breaks results down by data-quality fault. References checked against a primary source or the DOI registry.

`verification/references.md, bibliography verified against primary sources`

RESEARCH QUESTIONS

Four questions, one study each, hypotheses written first

RQ1	**Rules, classical and neural detectors in one pipeline** examples, 100 labelled problems	historical ranking, train-only replay pending
RQ2	**Which anomaly type each family finds** synthetic matrix, schemas, 42 UCI tables	narrow competence; values do not transfer
RQ3	**Operating point and time** contamination, label-free threshold, chronological split	a threshold fixed on the past fails when prevalence moves
RQ4	**Cost in the pipeline** cost study, ETL run on 152 sets	neural 10 to 100 times column-wise

Every study has a protocol and hypotheses committed before its runs. A hypothesis that fails is reported as failed: three of the six on the whole pool do.

`paper, Introduction and Methods; pre-registered in verification/*.md`

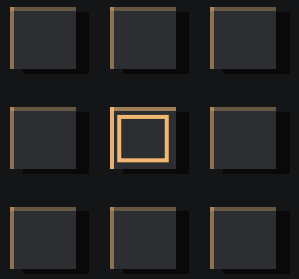
THE ORIGINAL CONTRIBUTION

A controlled dataset that labels six anomaly types separately



missing value

one or two feature cells set to null



exact duplicate

a clean row copied, same business attributes



univariate outlier

one numeric field pushed far out of range



multivariate outlier

every field in range, the combination not



temporal drift

recent signups whose income and balance shifted up



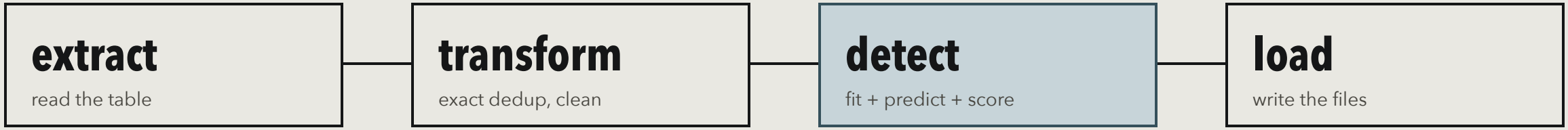
invalid format

malformed e-mail, bad country code, impossible age

12,360 records, Faker and NumPy, six types injected at 3% each, every row tagged. Recall and precision are reported per type. Severity is one multiplier, so the magnitude is swept rather than fixed. The glyphs show a pad missing, doubled, raised, shifted, sloping or cracked.

`make synthetic` · `make severity` · `generator is deterministic, dataset not committed`

One interface, one measured path



the stage whose latency is measured, in real in-pipeline time

The demo and every benchmark call the same `detect()` stage. A continuous `score()` on top of the boolean `predict()` makes ROC-AUC and AP computable; `detect(fit_df=...)` fits on the train slice and scores the test set.

Exact deduplication belongs to the ordinary transform. Typed injected duplicates remain in their separate evaluation with transform_dedup.

```
# every detector, one contract
class Detector(ABC):
    name: str
    def fit(self, df) -> Self: ...
    def predict(self, df) -> Series:  # bool/row
        ...
    def score(self, df) -> Series:    # graded
        return self.predict(df).astype(float)

# classical ML: adapt PyOD, do not reimplement
class PyODDetector(Detector): ...

# registry: one line per detector
"isolation_forest": lambda: IsolationForestDetector(),
```

METRIC INTERPRETATION

Average Precision measures the ranking

$$AP = \sum \Delta R \cdot P$$

precision, weighted by each recall increment, over distinct score cutoffs from high to low; tied scores enter together. AP uses continuous anomaly scores. Precision, recall, F1 and the flagged share use a selected threshold.

Non-interpolated

AP differs from trapezoidal precision-recall area.

Prevalence matters

Prevalence is the no-skill reference. Constant scores give AP equal to prevalence.

Decisions apart from scores

Changing only the flag threshold leaves AP and ROC-AUC unchanged; neither fixes a useful operating point.

Historical naming: stored pr_auc fields and remaining AP labels in archived figures and the paper denote this same AP quantity.

THREE FAMILIES, ELEVEN ROW-LEVEL DETECTORS

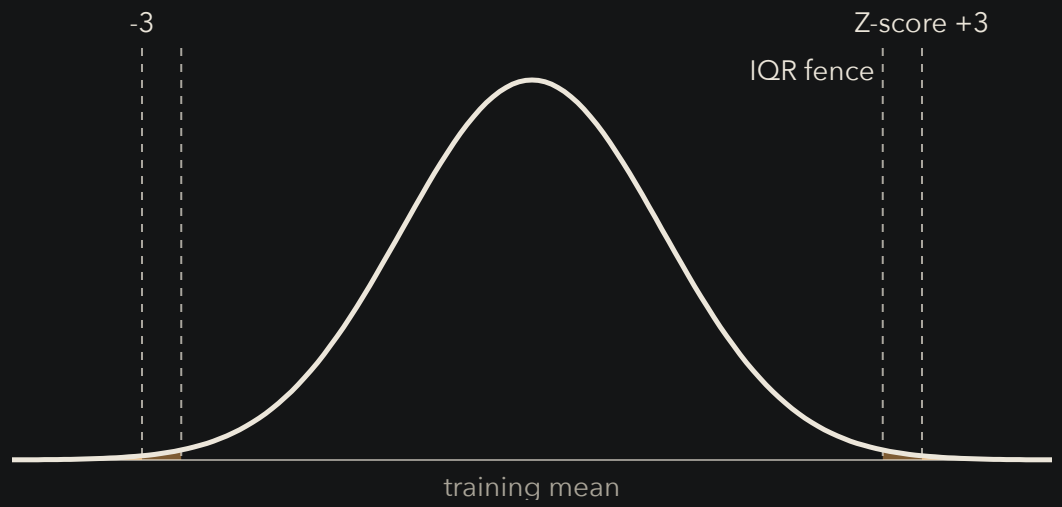
Rules and classical models, one contract

rules	**Z-score** flag |z| > 3 on any column limit: mean and std inflated by outliers	**IQR** Tukey fences, k = 1.5 limit: fires often in high dimension	**Great Expectations** null checks, fitted 0.1% / 99.9% bounds limit: generic suite, no domain rules		
joint models	**Isolation Forest** isolation depth under random splits limit: weak on nulls	**LOF** local density, 20 neighbours limit: cost grows with scored set	**One-Class SVM** RBF boundary around the bulk limit: quadratic training	**KNN** distance to the 5th neighbour limit: skipped above 50k rows	**PCA** reconstruction error, 95% variance limit: misses shifts inside the kept space
column-wise	**ECOD** empirical tail probabilities, summed limit: no cross-feature dependence	**COPOD** max of skew term and mean log tails limit: same	**HBOS** histogram per column, inverse bin height limit: independent columns, coarse bins		
time-aware	**drift** recent-window mean shift limit: per row, so diluted drift is invisible				

PyOD or the recorded protocol adaptations; flag threshold from training prevalence. Numeric columns only, while Great Expectations separately checks nulls on all columns: an explicit information asymmetry. The time-aware drift detector is opt-in and synthetic only.

`src/etl_benchmark/quality/ml.py, rules.py, gx.py · ECOD/COPOD use training ranks and frozen skew signs`

Column bounds do not fix row alert rates



Per-column tails under a Gaussian: **0.27%** beyond 3σ, **0.70%** beyond the IQR fences (about 2.7σ). Fitted bounds and several columns change row-level rates.

Shaded: the region a rule flags. Gaussian reference only.

Z-score

Max |x − training mean| / sample standard deviation. Flag above 3; constant columns contribute zero.

IQR

Training fences Q1 − 1.5 IQR and Q3 + 1.5 IQR. Flag strictly outside; zero-IQR columns contribute zero.

Great Expectations

Not-null on all columns; inclusive training quantiles 0.1% and 99.9%. Score: fraction of active checks violated. A generic statistical baseline: no domain-specific category, format or cross-field rules.

SIX NEURAL MODELS, OPT-IN

Neural detectors behind the same contract

Autoencoder	rebuild standardized rows through a bottleneck	MLP from scikit-learn
VAE	Gaussian code, rebuild, reconstruction error	scored with the code mean; PyOD draws one
Deep SVDD	map normal rows near a centre, distance	weakest on most types
DIF	isolation forests on random neural features	training representation scaler reused
AE1SVM	autoencoder + random-Fourier one-class SVM	singleton padding changes training weight
LUNAR	network learns the score from neighbour distances	best on drift, many false alarms

PyOD 3.6.7 with recorded adaptations, CPU, seeded, fitted on at most 20,000 rows.

Frozen representation scaling and latent-mean scoring change the evaluated algorithms. They do not establish equivalence to upstream defaults.

The LSTM autoencoder reads windows of ordered rows and stays outside the registry.

EVALUATED VARIANTS

Stable row scores require explicit adaptations

A fitted ETL detector must score a row without consulting other evaluation rows. Contract tests check the 17 row-level detectors; the time-window monitor has a different contract.

EVIDENCE BOUNDARY

Bounded tests support batch independence, not universal equivalence. Separate ablations remain open; historical results keep their actual adapted configurations.

ECOD and COPOD

PyOD 3.6.7 recomputes tails and skew with the evaluated batch. Here ranks and skew signs are frozen from training.

Copula motivation versus score

COPOD adds marginal contributions: it is not a fitted general joint copula. ECOD takes the max of both log tails and the skew term.

Neural scoring

VAE decodes the latent mean; DIF reuses a training scaler; AE1SVM padding reweights its last row. These are algorithm changes.

EXPLORATORY CORRECTION NOT A REPLACEMENT OF ARCHIVED RESULTS

Historical pool preparation needs revalidation

Historical path

Column filtering, category vocabulary and conflicting-label removal happen **before** the split. Training-only medians and scaling do not isolate those earlier decisions.

Constructed negative controls: changing only the evaluation categories changes the encoded width from 5 to 3; an evaluation duplicate's label can remove a reference row. The effect on AP is not yet quantified.

pool-train-only-v1

Consumes source-bound frozen positions. Fitted schema, imputation and scaling use training rows only. Cross-partition duplicates are rejected; evaluation rows are retained. Targets are excluded from preparation.

Bounded checks enforce full line and branch coverage for the two new modules and detect three leakage defects.

STILL OUTSTANDING

Historical memberships were recomputed (not originally archived) and replay-validated: the Z-score reproduces all 100 archived per-seed results. **Replaying the studies on the train-only path remains outstanding.** Old protocols and results are preserved, and no corrected pool-performance claim is available.

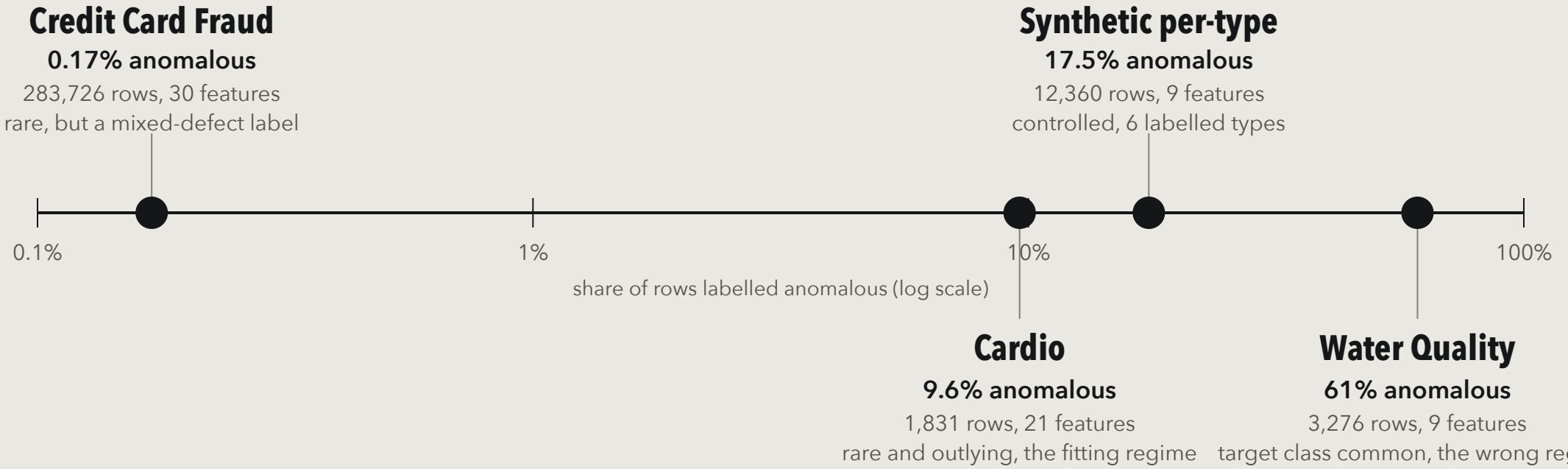
What the detectors actually find

Three example datasets, then six anomaly types on a controlled generator and on real tables.

Illustration: the six defect types drawn on one plate (coded SVG, not a photograph, not data).

THREE EXAMPLES, ONE SYNTHETIC SET

Three example datasets bracket the method



Stratified 70 / 15 / 15 split, fit on train, every number reported on the held-out test set. Five seeds, mean ± std. The ML contamination is read from the train-set prevalence, not fixed at 10%.

LOF and One-Class SVM run everywhere except Credit Card Fraud, where neither scales to 284k rows in the pipeline.

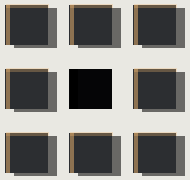
0.01 → 0.20

Isolation Forest fraud precision, from the contamination change alone

`src/etl_benchmark/datasets.py` · `config.BENCHMARK_SEEDS` · `data/*.csv gitignored`

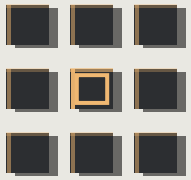
HOW EACH TYPE IS INJECTED

The magnitudes are a swept parameter, not a fixed choice



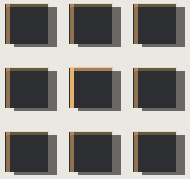
missing value

1 to 2 of {income, balance, avg amount, tenure, n txn, email, country} set to null



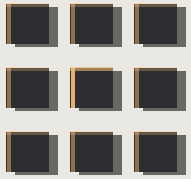
duplicate

exact copy of a sampled clean row, id included (appended, not replaced)



univariate

one numeric column × U(15, 45) × severity



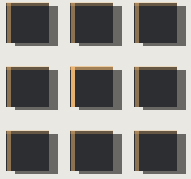
multivariate

marginals at the (1 − 0.03 / severity) percentile of the clean data



temporal drift

signup in the last 80 days; income × (1 + 0.55 · severity), similar for balance and amount



invalid format

email malformed; age impossible; country out of domain

Severity 1.0 is the reference dataset. Missing values, duplicates and invalid formats have no magnitude to scale. Five types replace clean rows; duplicates are appended. Sweeping the multiplier turns "we picked the magnitudes" into a curve (severity slide), and the per-type ranking does not change.

Aggregate scores do not establish equivalence.

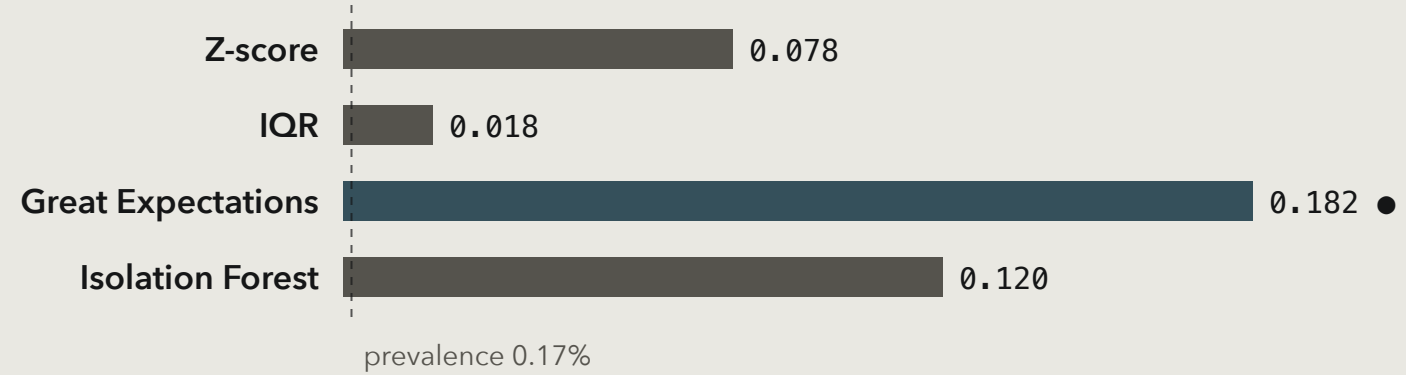
Per-type results show different competencies.

F1 spans 0.22 to 0.59 over the seventeen detectors on the synthetic set, and the per-type competences barely overlap.

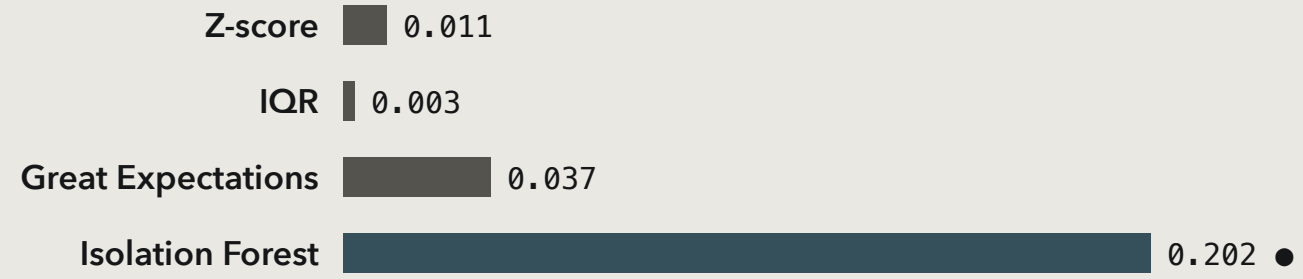
REAL FRAUD DATA

The operating point decides the outcome

AVERAGE PRECISION (RANKING)



PRECISION AT THE FLAG THRESHOLD (DECISION)



● best of its chart. Mean over 5 seeds, 15% held-out test split. LOF and One-Class SVM do not scale to 284k rows here.

The statistical rules recover most frauds (recall 0.87 and 0.97) at precision below 0.04. Isolation Forest changes once its contamination matches the prevalence: it flags 0.17% of rows and reaches the first usable precision, at recall 0.21 (F1 = 0.20 ± 0.07). Great Expectations leads AP, at precision 0.04.

0.01 → 0.20

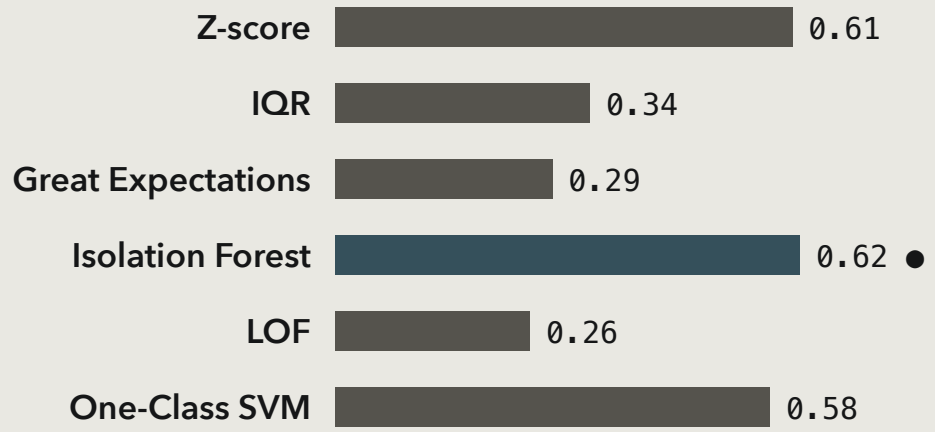
Isolation Forest precision when contamination goes from 0.10 to the prevalence

`experiments/run_benchmark.py · make benchmark · the metric that matters here: AP`

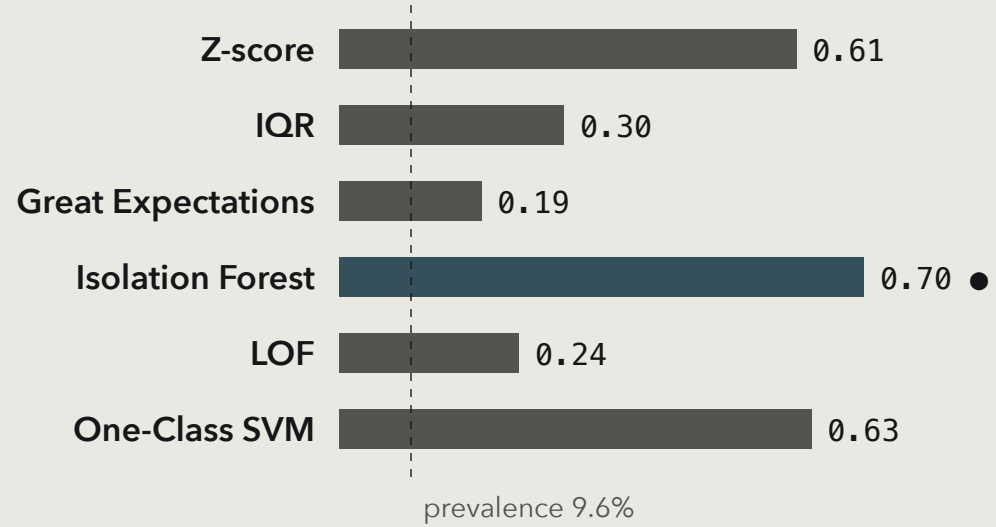
GENUINELY IMBALANCED REAL DATA

When anomalies are rare and outlying, the learned detectors work

F1 AT THE FLAG THRESHOLD



AVERAGE PRECISION

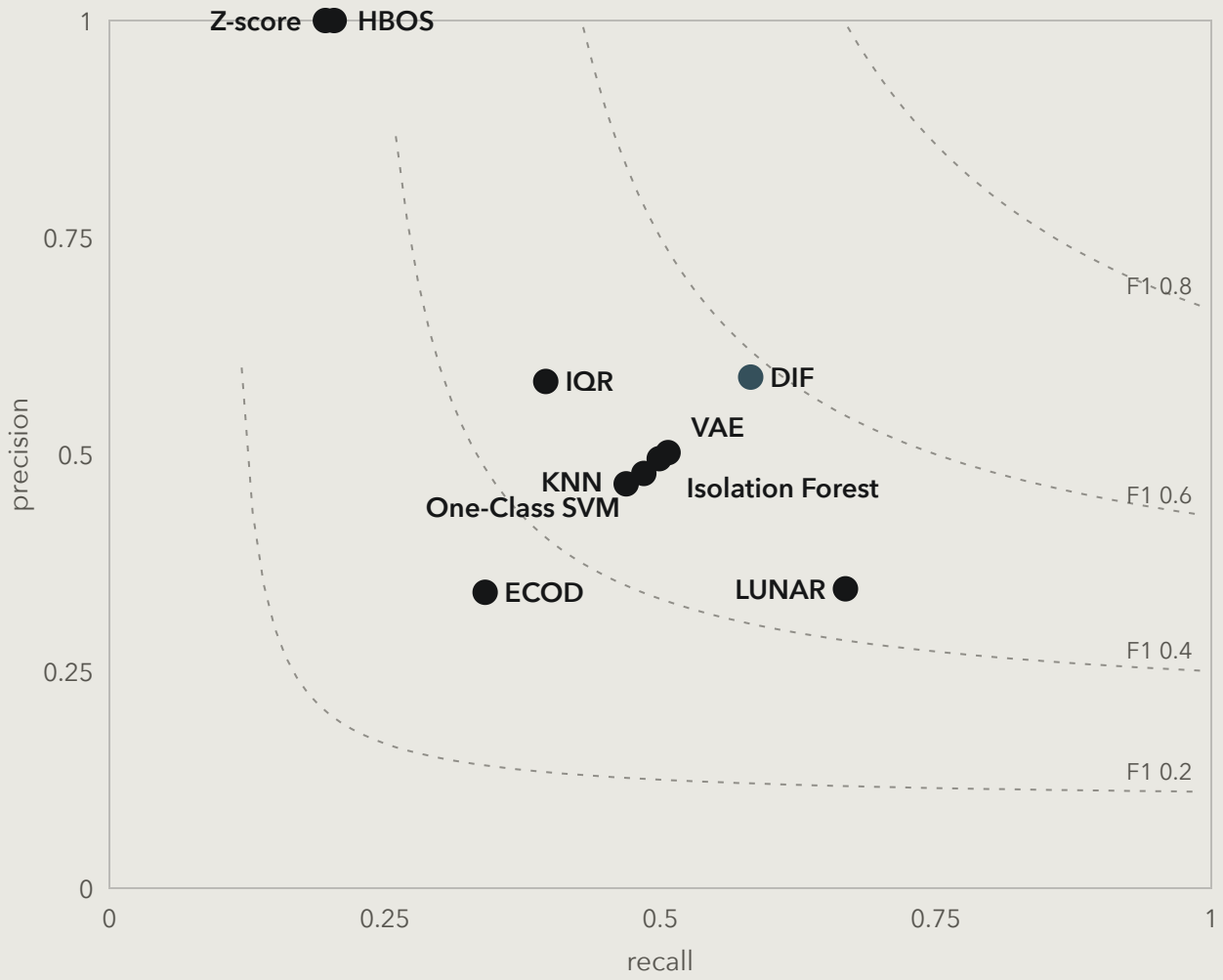


Isolation Forest reaches F1 0.62 and AP 0.70, with ECOD, COPOD and DIF close behind (AP 0.69, 0.69, 0.67). Z-score also does well: the 21 features are standardized and anomalies stand out on several at once. This is the counterpoint to a 0.17% mixed label and a 61% majority.

● best of its chart. Cardiotocography, 1,831 rows, 9.6% anomalies. Mean over 5 seeds, test split. Wide seed spread on this small set (ML F1 std up to 0.17).

SYNTHETIC DATASET, AGGREGATE

Aggregate scores hide different trade-offs



Z-score and HBOS reach a perfect precision by firing only on extreme rows (recall 0.20). DIF has the best F1 and AP; the VAE matches Isolation Forest; LUNAR has the best recall with a low precision.

F1 0.22 to 0.59

range across the seventeen detectors

Ten of the seventeen detectors shown, mean over 5 seeds, test split, 12,360 rows, 17.5% anomalous. Dashed lines are constant F1. ● best F1. All 17 are in the paper, table tab:synthetic.

verification/evidence/full_registry · paper, tab:synthetic

PER TYPE, THE CONTRIBUTION

Each method has a narrow competence

Z-score owns univariate outliers (0.93). **Great Expectations** owns missing values (1.00). **Deduplication** owns exact duplicates.

Isolation Forest (0.89) and the One-Class SVM (0.90) are the only first-six methods with real recall on multivariate outliers; every rule stays at 0.04 or below.

Broad coverage needs several families, not the single best one.

	missing	duplicate	univariate	multivariate	drift	invalid
Z-score	0.00	0.00	0.93	0.00	0.00	0.24
IQR	0.05	0.04	0.99	0.04	0.25	1.00
GE	1.00	0.00	0.21	0.00	0.00	0.00
IF	0.09	0.11	0.99	0.89	0.33	0.60
LOF	0.28	0.16	0.70	0.21	0.50	0.67
OCSVM	0.07	0.10	1.00	0.90	0.29	0.54
ECOD	0.10	0.13	0.68	0.52	0.26	0.36
COPOD	0.09	0.13	0.89	0.51	0.34	0.30
HBOS	0.00	0.00	0.90	0.00	0.00	0.32
KNN	0.09	0.10	1.00	0.39	0.41	0.81
PCA	0.10	0.12	1.00	0.48	0.23	0.58
AE	0.12	0.10	0.81	0.83	0.29	0.53
VAE	0.07	0.11	0.99	0.92	0.30	0.66
DSVDD	0.19	0.18	0.33	0.13	0.16	0.32
DIF	0.09	0.07	1.00	1.00	0.33	1.00
AE1SVM	0.09	0.10	0.99	0.78	0.27	0.61
LUNAR	0.44	0.39	1.00	0.63	0.65	0.90

0 — 1 (recall)

Mean recall, 17 detectors, synthetic customer table, 5 seeds. Colour scale: cividis, 0 to 1.

paper, Appendix C, tab:m-customer · duplicates need the transform stage

RECALL AND PRECISION, PER TYPE

Detection rate alone can be **gamed**

A wide net

0.89 recall

~0.22 precision

Isolation Forest and the One-Class SVM on multivariate outliers: they cast a wide net that also catches clean rows.

A narrow one

1.00 precision

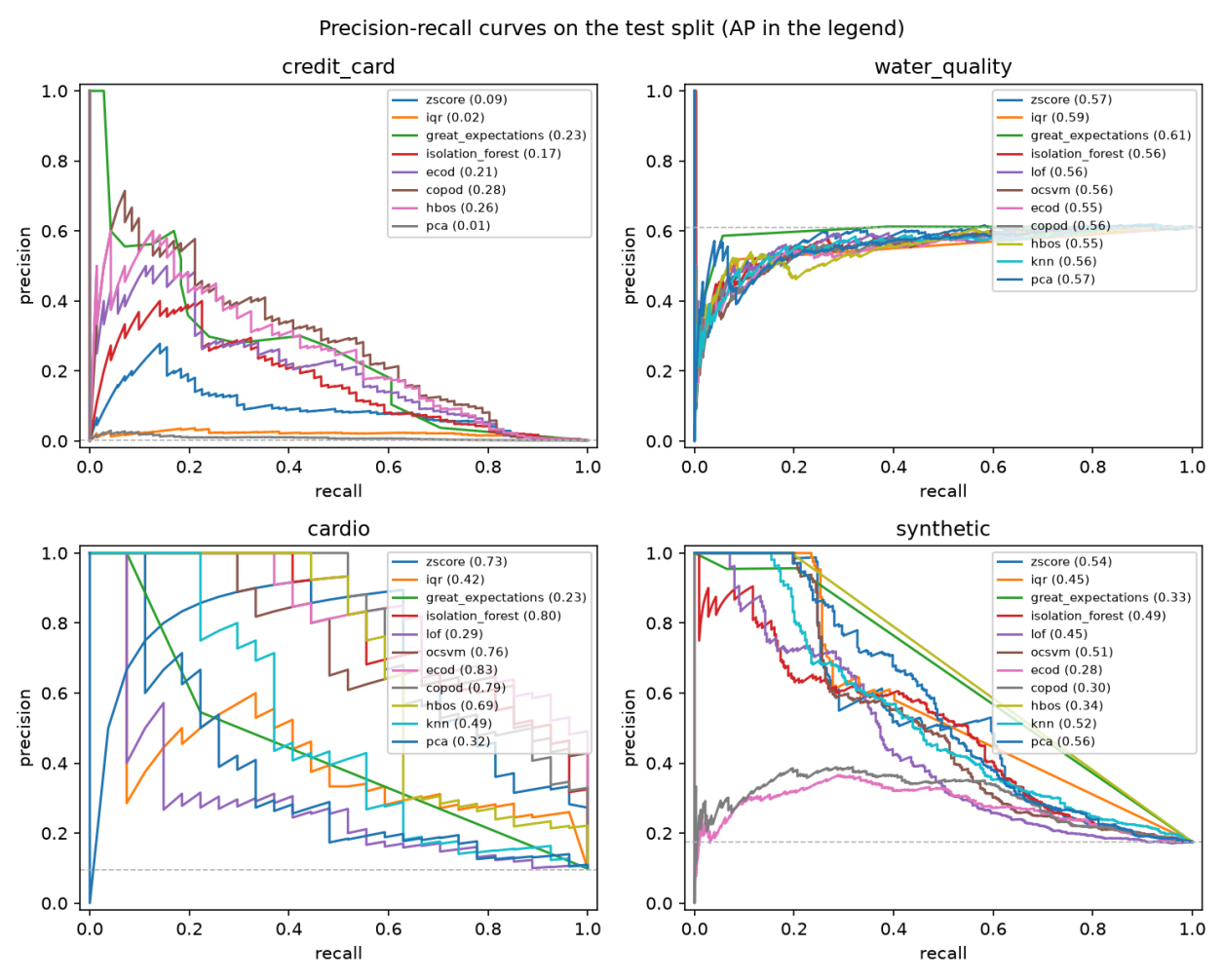
0.20 recall

Z-score fires only when it is certain: on the whole synthetic set its precision is 1.00 and its recall 0.196.

Clean rows are the only negatives. Both views are needed: neither number alone ranks the detectors.

`paper, per-type results and tab:synthetic · synthetic customer table, 5 seeds`

One threshold hides the whole trade-off



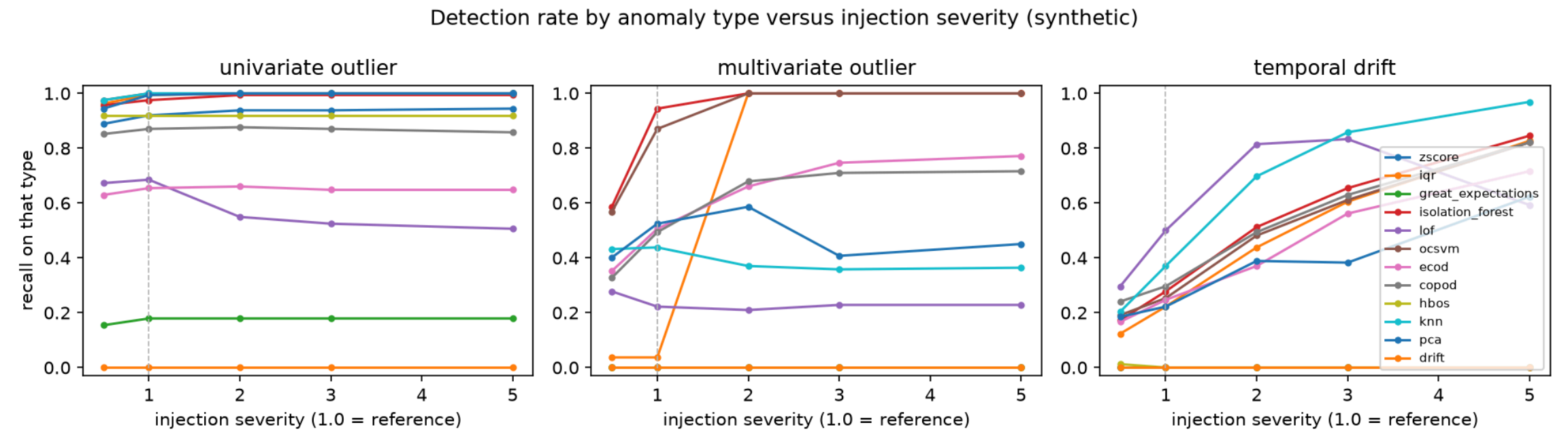
On **Cardio** the Isolation Forest and One-Class SVM curves stay high (AP 0.80, 0.76). On **Water Quality** every curve hugs the 0.61 base-rate line. On **Credit Card Fraud** two methods dominate but neither clears precision 0.4 except at negligible recall.

0.37 → 0.008

Isolation Forest fraud precision as recall goes 0.23 to 0.92

experiments/run_curves.py · make curves · archived figure from paper/figures

The per-type ranking is stable as the magnitudes scale



Univariate outliers are saturated already at severity 0.5. Multivariate outliers show a threshold. Temporal drift climbs roughly linearly for the distributional detectors: the reference drift is below every method's floor, not a property of the methods. The same families lead the same columns at every severity.

≤0.3 → 0.85

temporal-drift recall from severity 1 to 3

From three examples to a pool of 152

Real tables, pinned and checked; what holds, what fails, and what changes with time.

Illustration: 152 pads, 100 of them with a lit edge, one for each labelled problem. Coded SVG, not data.

A pool of 152 open datasets, pinned and checked

ADBENCH

47 tabular sets in the commit → **36** kept: terms read and open

11 excluded: no open licence or terms forbid use (celeba, mammography, vowels ...)

UCI REPOSITORY

689 catalogue entries → **133** pass metadata and licence filters → **116** kept after curation

17 removed with a reason (version duplicates, rule-labelled, boards)

36 + 116

= **152** pinned datasets: ADBench kept plus UCI curated

116

UCI tables as clean carriers for typed injection

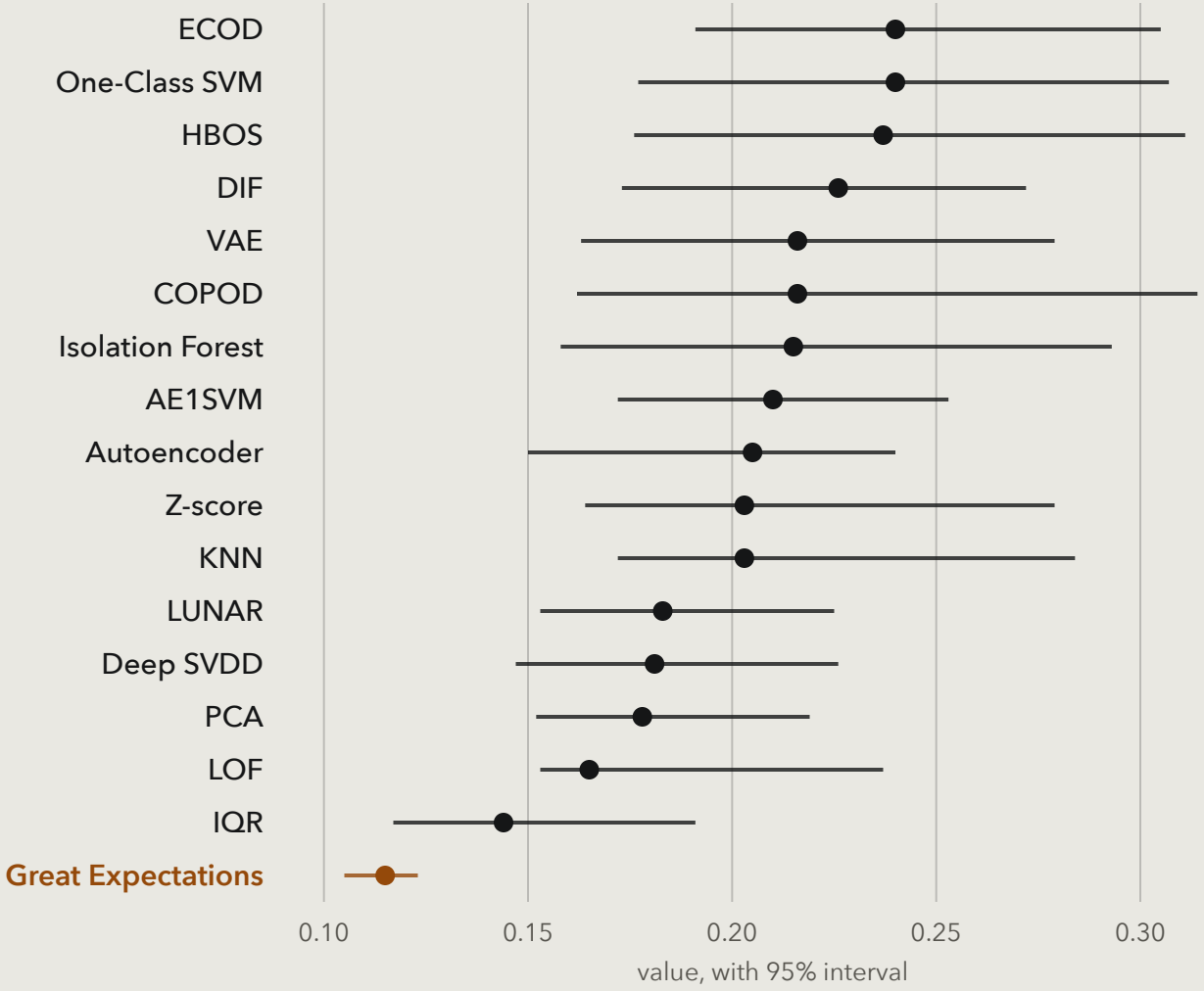
36 + 64

= **100** labelled problems; rule fixed before any detector: the rarest class is the anomaly

One normalisation policy for all sets, applied in order. 19 sources appear in both lists with a different derivation. Raw data stays out of the repository; checksums, licences and reports are archived. A download check of every set runs manually and monthly.

HISTORICAL RUN, NEEDS TRAIN-ONLY REPLAY

Historical pool rankings need train-only revalidation



Median AP, 95% interval, 100 problems. Intervals overlap: the order is descriptive. LOF, KNN and One-Class SVM skip large problems (92, 92, 84).

Three rules, eight classical and six neural detectors, five seeds, on 36 ADBench and 64 UCI problems. The numbers are retained as recorded.

OPEN Schema and row filtering precede the split, so these descriptive rankings still need train-only replay.

The 100 problems come from 80 sources; a source-clustered re-analysis changes no sign (only Great Expectations, in amber, is clearly below Isolation Forest). Label-rule sensitivity and unrecorded PyOD convergence diagnostics remain open.

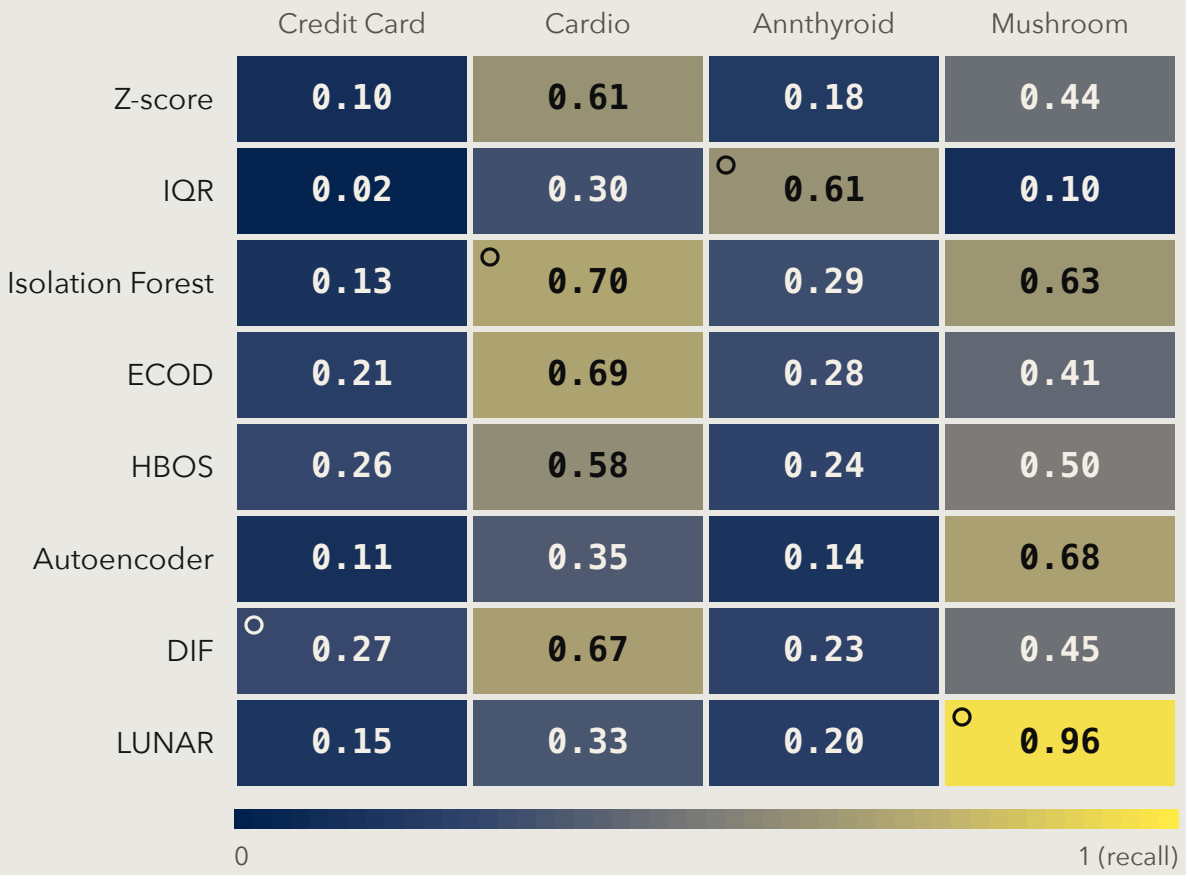
17

detectors; no neural exception

13

problems won by DIF, the most

Two regimes the three examples do not show



Among the 100 problems, the largest margin of the rules over the learned models is on **Annthyroid**: IQR 0.61, best learned 0.31. The largest margin of a neural model is on **Mushroom**: LUNAR 0.96, best classical 0.63 (label derived by a rule), where the IQR rule is at chance, 0.10.

Chosen after the results, as illustrations, not as a test.

Average precision, mean over 5 seeds, as run in the pool. Circle: best of its column. Colour: cividis, 0 to 1. All 17 detectors are in the paper.

The synthetic pattern does not simply carry over

SYNTHETIC CUSTOMERS, MEAN RECALL

	miss.	dup.	univ.	multi.	drift	inv.
Z-score	0.00	0.00	0.93	0.00	0.00	0.24
IQR	0.05	0.04	0.99	0.04	0.25	1.00
GE	1.00	0.00	0.21	0.00	0.00	0.00
IF	0.09	0.11	0.99	0.89	0.33	0.60
LOF	0.28	0.16	0.70	0.21	0.50	0.67
OCSVM	0.07	0.10	1.00	0.90	0.29	0.54
ECOD	0.10	0.13	0.68	0.52	0.26	0.36
COPOD	0.09	0.13	0.89	0.51	0.34	0.30
HBOS	0.00	0.00	0.90	0.00	0.00	0.32
KNN	0.09	0.10	1.00	0.39	0.41	0.81
PCA	0.10	0.12	1.00	0.48	0.23	0.58
AE	0.12	0.10	0.81	0.83	0.29	0.53
VAE	0.07	0.11	0.99	0.92	0.30	0.66
DSVDD	0.19	0.18	0.33	0.13	0.16	0.32
DIF	0.09	0.07	1.00	1.00	0.33	1.00
AE1SVM	0.09	0.10	0.99	0.78	0.27	0.61
LUNAR	0.44	0.39	1.00	0.63	0.65	0.90

42 REAL UCI TABLES, MEDIAN RECALL

	miss.	dup.	univ.	multi.	drift	inv.
Z-score	0.00	0.00	0.00	0.00	0.00	0.43
IQR	0.19	0.21	1.00	0.39	0.64	0.93
GE	1.00	0.00	0.23	0.00	0.00	0.42
IF	0.11	0.13	0.61	0.37	0.45	0.59
LOF	0.28	0.13	0.68	0.33	0.34	0.70
OCSVM	0.10	0.13	0.75	0.17	0.31	0.73
ECOD	0.10	0.14	0.44	0.25	0.36	0.44
COPOD	0.10	0.15	0.45	0.23	0.48	0.41
HBOS	0.00	0.01	0.04	0.02	0.01	0.32
KNN	0.19	0.10	0.91	0.30	0.46	0.80
PCA	0.12	0.14	0.54	0.19	0.23	0.71
AE	0.11	0.12	0.76	0.29	0.36	0.76
VAE	0.10	0.13	0.72	0.18	0.27	0.72
DSVDD	0.14	0.13	0.72	0.23	0.37	0.70
DIF	0.11	0.12	0.81	0.28	0.39	0.78
AE1SVM	0.11	0.14	0.63	0.17	0.18	0.76
LUNAR	0.38	0.25	0.98	0.63	0.78	0.90

0 — 1 (recall)

Two rows hold everywhere: Great Expectations finds all missing values, deduplication all duplicates. **Three of five hypotheses are not supported.**

0.89 → 0.37

Isolation Forest, multivariate: synthetic to real

24%

median clean rows flagged by LUNAR

Injected, not natural anomalies; protocol and five hypotheses written before any run. Same row order and the same cividis scale (0 to 1) in both matrices. Real-table rows from verification/evidence/full_registry/summary.md. The duplicate and drift detectors are omitted here.

READING THE TYPED RESULT

Two failures that are explained, not hidden

0.00

Z-score, univariate outliers

Median 0.00 against 0.93 synthetic. A check made after seeing it, so exploratory: invalid rows in the training split inflate the standard deviation (**masking**). Without them the Z-score finds every univariate outlier on all 42 tables.

22%

clean rows flagged by IQR

The IQR fences reach the best detection rate on several types, but only by flagging a median 22% of clean rows; the false positive rate reaches 99% on some wide tables. Detection rate has to be read next to it.

0.83

mean rank correlation between two tables (0.70 for drift, partly mechanical)

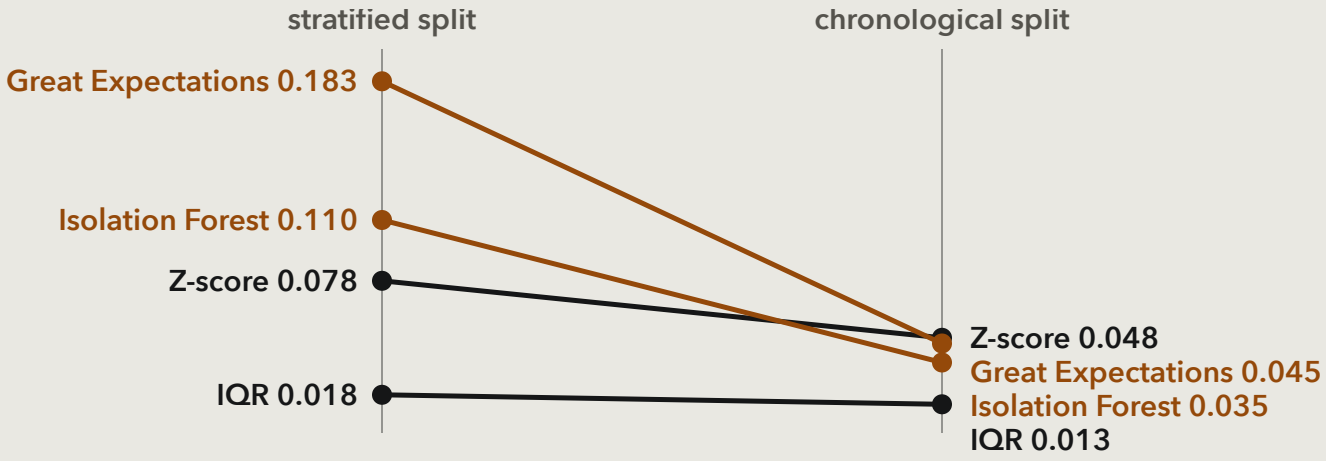
−0.96

Isolation Forest univariate detection against the number of columns

Injected anomalies, one severity. Deep rows are added on a later slide.

CHRONOLOGICAL SPLIT

Time changes the answer: a threshold from training does not travel



Average precision on the test split. Amber: the 95% block-bootstrap interval excludes the stratified mean (Great Expectations, Isolation Forest). Scale 0 to 0.2.

0.936

Isolation Forest ROC-AUC, chronological

0.08%

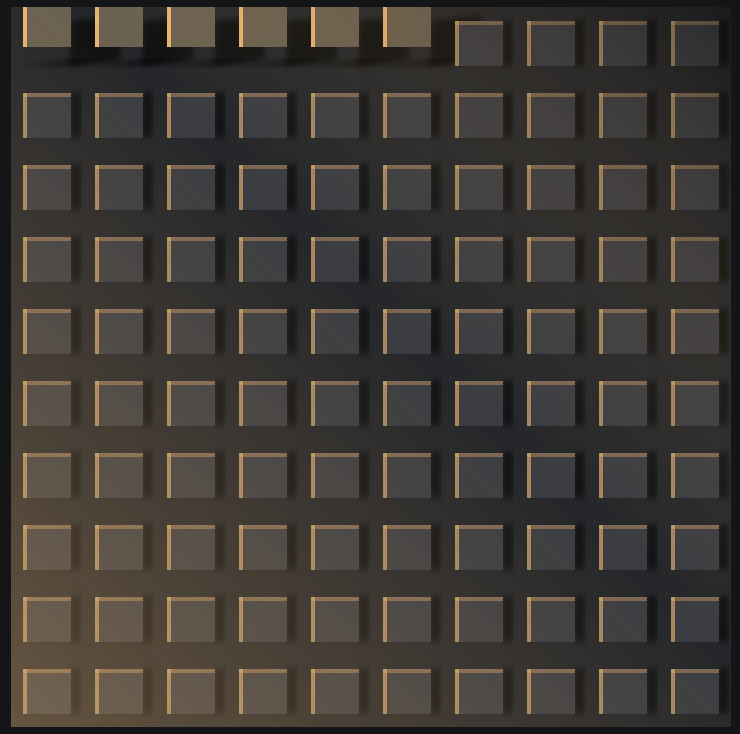
future rows it flags: none of the 52 frauds

Credit Card is the only dataset with a clock. Detectors are fitted on the first 70% of time and scored on the last 15% (52 frauds). Average precision drops for all four detectors.

CAVEAT Prevalence also changes between partitions, so the pilot does not separate it from the effect of time.

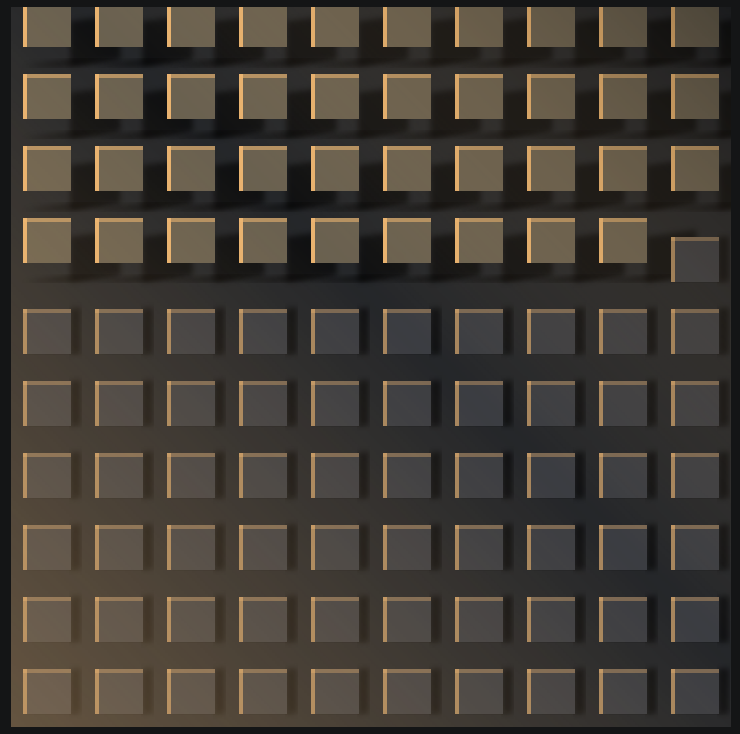
The window monitor detects mean shifts of 1 and 2 standard deviations, not 0.5.

Part of the gap is only the prevalence



5.8%

anomalous in training, Bank Marketing



39.3%

anomalous in test, same dataset

Illustration: each grid is 100 pads; raised pads stand for the anomalous share (6 and 39 pads, rounded).

Credit Card and Bank Marketing (UCI, ordered by date 2008 to 2010), four detectors, three label-free thresholds. Matching the stratified control to the test prevalence removes most of the gap. A rolling threshold that keeps the training prevalence cannot follow it.

One hypothesis of four holds.

0.26 → 0.06

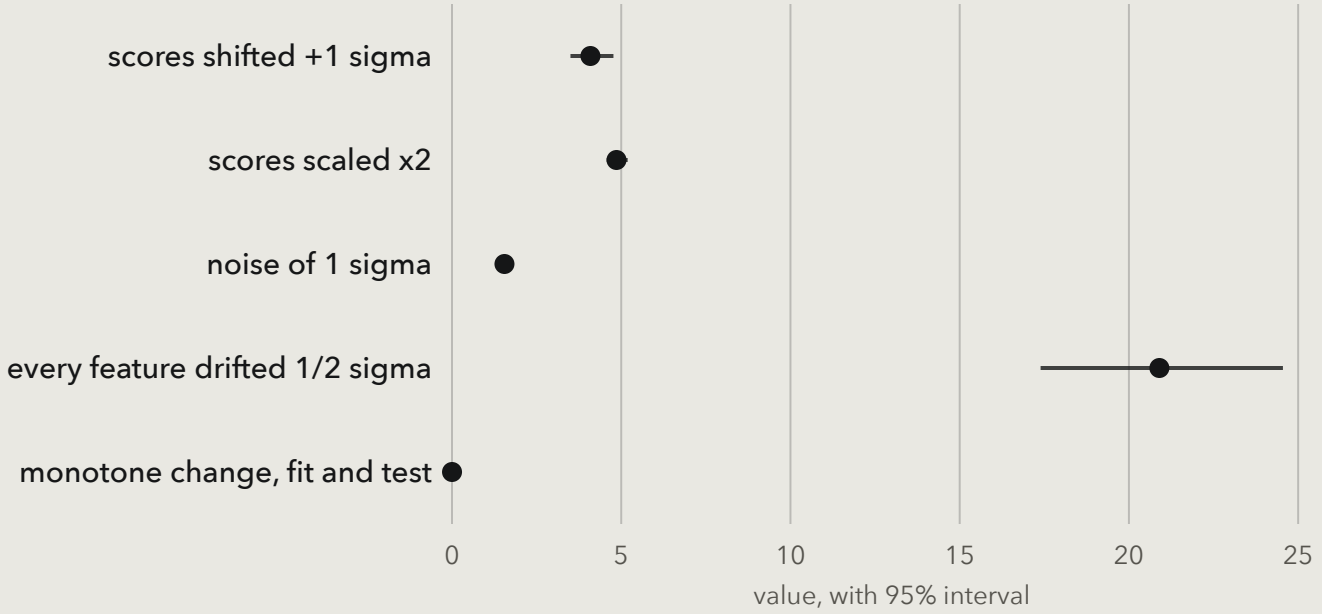
median AP gap on Bank Marketing after prevalence matching

F1 at the threshold, test period: Bank Z-score 0.51 (P0) against 0.02 (P2 rolling); Isolation Forest 0.41 against 0.11. Rank correlation of the detector order, stratified against chronological: 0.80 on Bank Marketing, 0.40 on Credit Card. Kyoto-2006+ left out: rights and download not verified.

 EXPLORATORY, ONE MACHINE

A nominal alert budget is not a promise

ADDED ALERT VOLUME AT A 2% BUDGET, IN BUDGETS (95% INTERVAL)



Five classical detectors, 98 problems from 78 sources, sources resampled whole, Holm-adjusted. Delta ROC-AUC is 0.000 for the shifts and scalings and the monotone change, -0.038 for noise and -0.051 for feature drift. Prevalence reweighted under a label-shift assumption: 0.83 to 1.83 times the budget at 0.5% prevalence, 1.15 to 2.73 times at 20%.

The 30-expected-alert rule for monitoring was chosen after the false alarms were seen. Not promoted to the main tables.

Budgets of 0.5 to 10% are set from training scores only, on 100 labelled problems. Held-out alert volume was **1.05 to 1.36** times the nominal 1% budget for classical detectors and **1.13 to 2.87** times for neural ones.

+4.1

budgets of added alert volume at 2% (about 5 times nominal) for a 1 sigma score shift, with ROC-AUC unchanged

EVERY STUDY ON THE WHOLE POOL

The deeper studies, run on all 152 datasets

P1	label-free threshold lowers F1	✓ HOLDS	14 of 17 detectors
P2	curve ranking = AP ranking	✗ NOT SUPPORTED	Spearman 0.90 (< 0.99)
P3	recall rises with severity	✗ NOT SUPPORTED	13 of 17; column-wise rules stay at 0
P4	detection share of the pipeline	✓ HOLDS	62% (IF), 4% (Z-score), 14 datasets
P5	chronological AP lower	✗ NOT SUPPORTED	31 of 64 pairs; Bank Marketing 0/16
P6	files match the manifest	✓ HOLDS	152 of 152

58 / 152

tables eligible for typed injection and severity

290 / 304

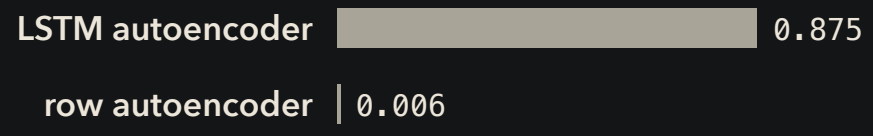
end-to-end pipeline runs that completed

Six hypotheses written first: three hold, three do not. Chronology needs a real time variable: 5 labelled datasets qualify (ADBench has none). Each study ran on every dataset where it is defined; the others are listed with a reason, never silently.

verification/pool_extension_results.md

An LSTM autoencoder is comparable, not clearly better

SYNTHETIC STREAM WITH BROKEN CONTINUITY, AP



On a synthetic stream with broken continuity the LSTM wins by a wide margin. The window is the ten previous rows of the global stream, since Credit Card has no cardholder. Protocol and a CPU budget of 20 epochs were fixed beforehand.

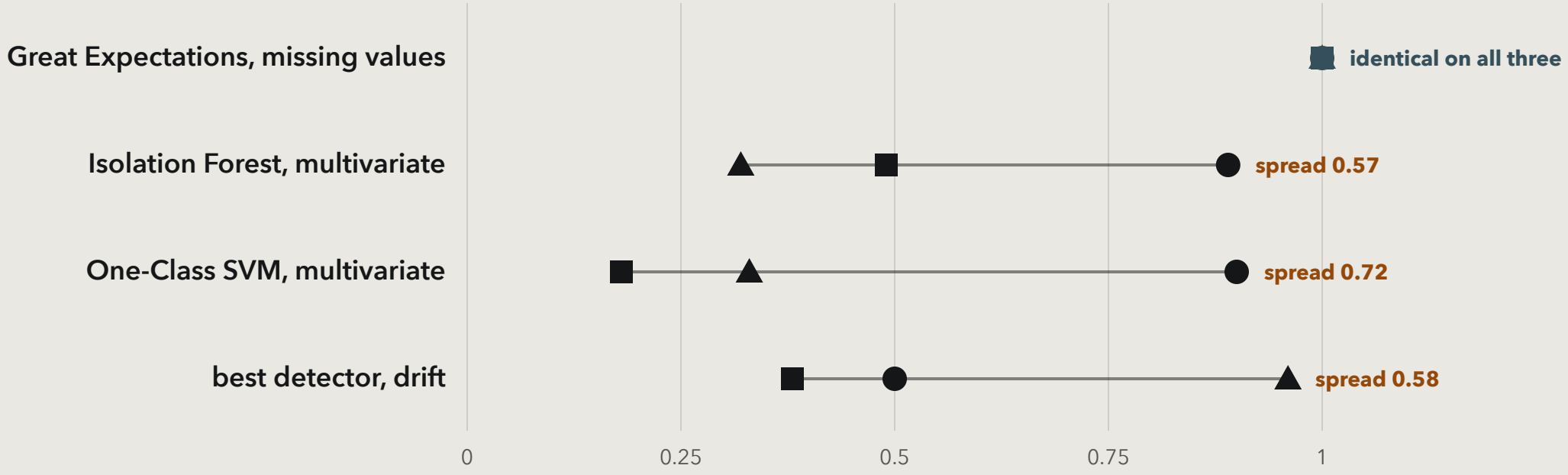
REAL CREDIT CARD FRAUDS, AP (NOTE THE SCALE)



On the real frauds it only matches Isolation Forest and Great Expectations. Paired bootstrap, main seed: +0.021 [+0.005, +0.044]; on the other four seeds the interval includes 0.

LIMIT None of five trainings converged within the 20-epoch budget. Recall of frauds in the later period is at most 0.02, and in the per-type matrix the LSTM does not fix drift.

The profile carries over, the values do not



● customers ■ transactions ▲ sensors. Mean recall over five seeds. Spread = highest minus lowest of the three.

Same generator logic for all three schemas: the agreement shows the profile does not depend on one table, not that it holds on real data. The drift detector flags nothing on any schema. Two of four hypotheses hold: the structural rows and the rank stability.

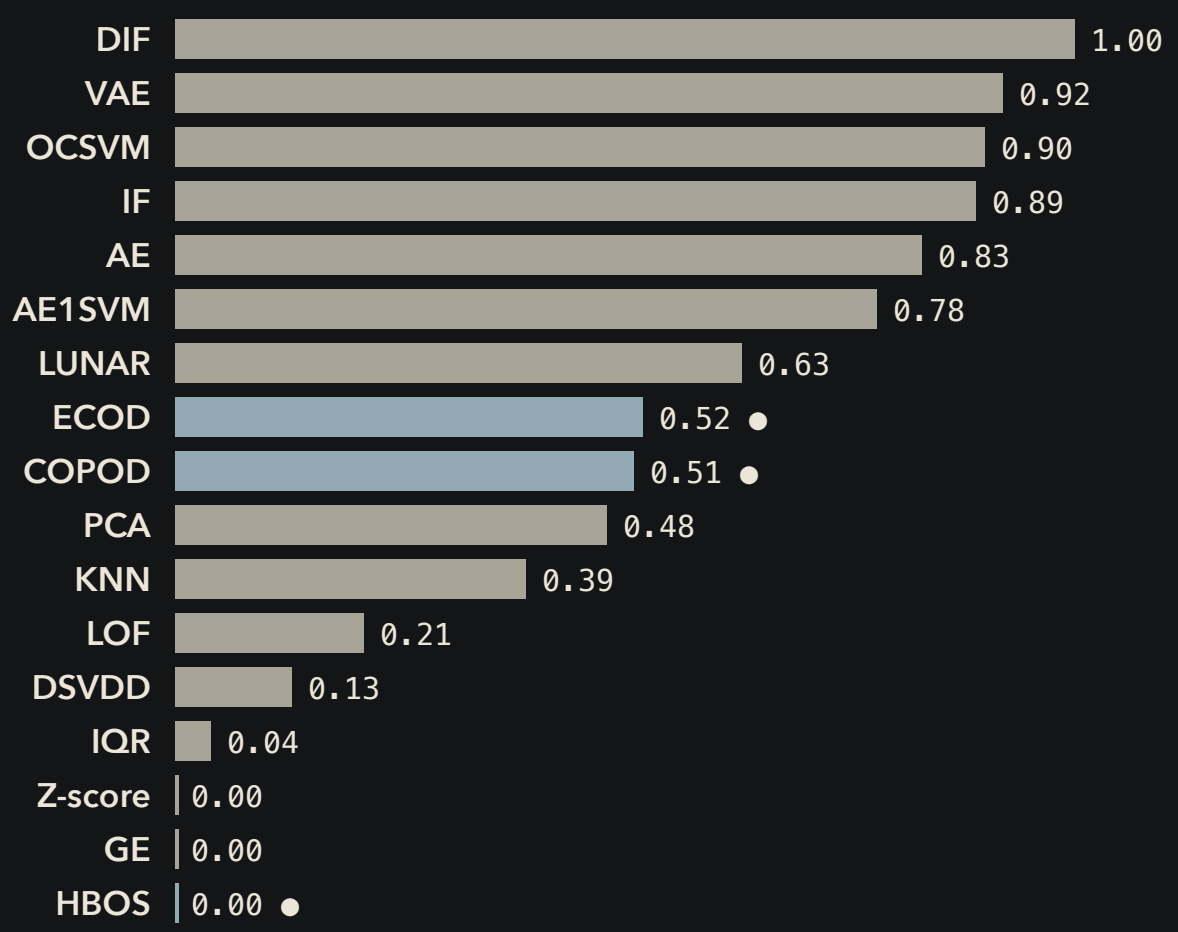
0.93

mean rank correlation between schemas, first six detectors (0.89 with all seventeen)

`verification/multi_schema.md` · `research/multi_schema.py` · paper, Section RQ2 transfer

EVERY DETECTOR IN THE PER-TYPE MATRIX

More detectors, same competence map, no hypothesis holds



Mean recall on multivariate outliers, 17 detectors, 5 seeds. Full matrix on the per-type slide.

Eleven rows added to the synthetic matrix, with three hypotheses written beforehand; **none holds**. Recall on multivariate outliers, sorted: ECOD (0.52) and COPOD (0.51) find half of them although they read one column at a time, and HBOS stays at 0; PCA reaches 0.48 against 0.83 for the autoencoder.

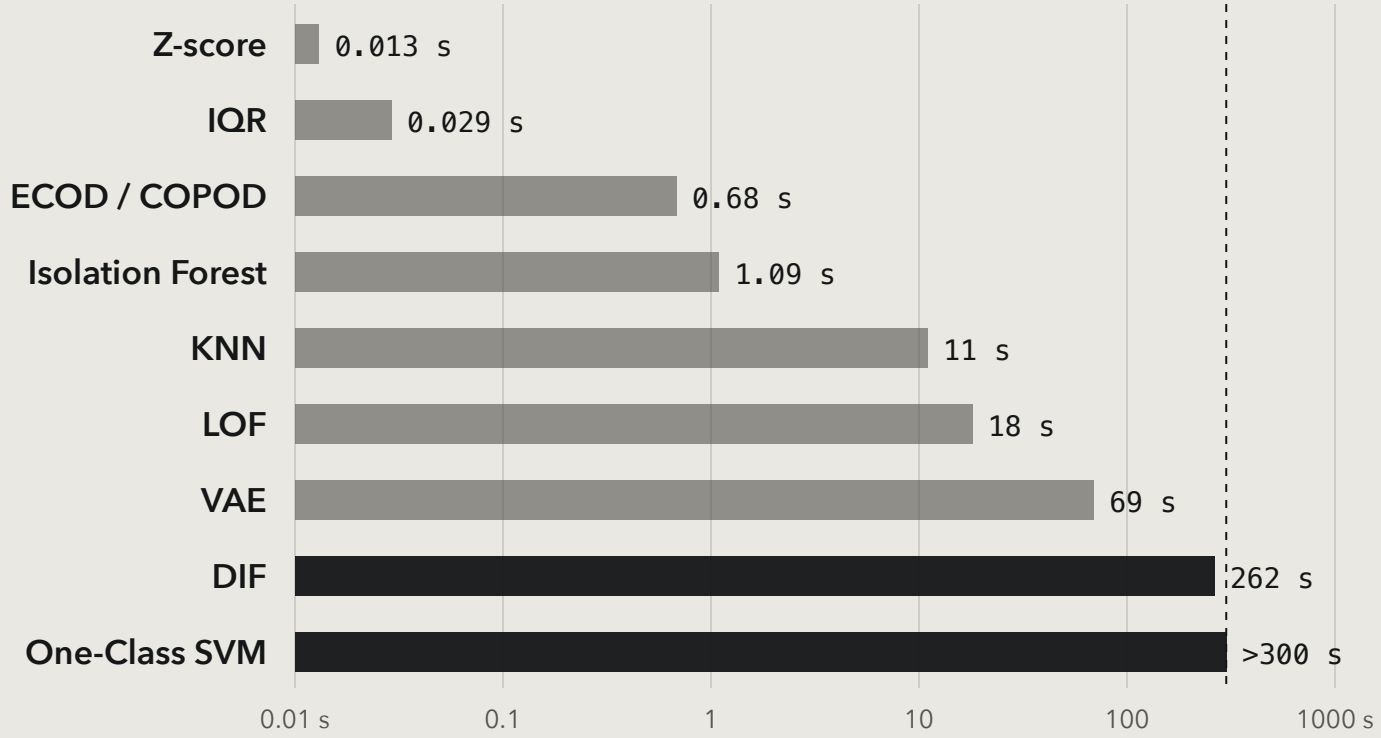
LUNAR catches 0.65 of the drifted rows (LOF 0.50) while flagging 41% of clean rows. The LSTM still does not fix drift.

■ column-wise detectors (ECOD, COPOD, HBOS), a family, not a result. Synthetic customers are independent rows: drift is not a sequence here.

`verification/full_registry.md · research/full_registry.py · paper, tab:m-customer`

Detection latency is not pipeline latency

MEDIAN DETECTION-STAGE TIME AT 100,000 ROWS, 8 FEATURES (LOG AXIS)



■ dark bars: over 100 s. Dashed: the 300 s budget. Fresh process per measurement so peak memory belongs to that run; synthetic matrices from 1,000 to 300,000 rows.

At 300,000 rows LOF takes 133 s (exponent 1.51) and DIF more than 300 s. Single machine, not isolated, not a general ranking. Five of the ten newer detectors shown; the paper has all.

On the full Credit Card file the Z-score pipeline takes **4.8 s**, of which detection is 3%. Isolation Forest takes **8.5 s**, with detection at 43 to 45%.

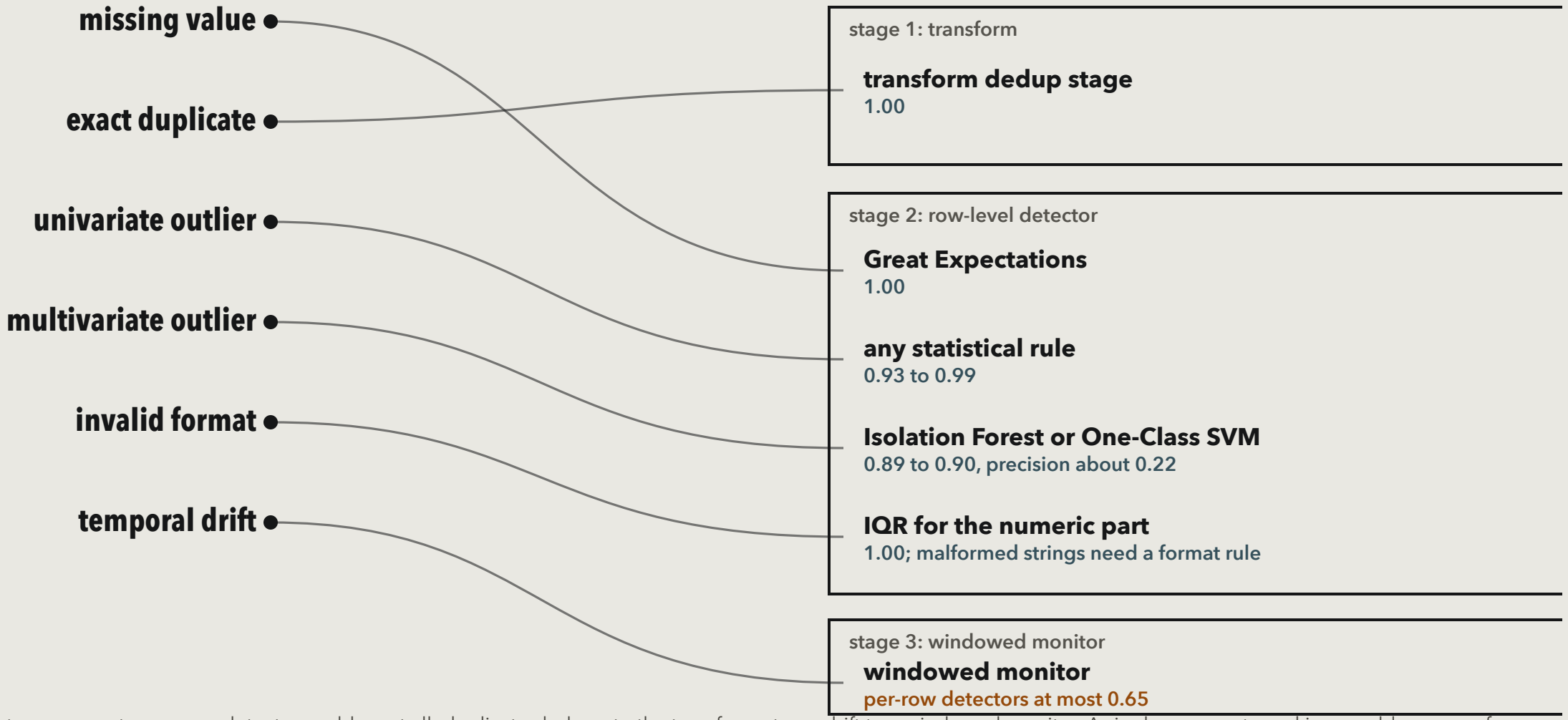
8.5 s

Isolation Forest pipeline, detection 43 to 45%

verification/cost_study.md

THE PRACTICAL OUTPUT

Anomaly type to method



Two of six types are not a per-row detector problem at all: duplicates belong to the transform stage, drift to a windowed monitor. A single aggregate ranking would never surface that. Recall from the synthetic customer table, 5 seeds; steel text: recall reached, amber: per-row detectors only partly succeed.

paper, Section V · rules and ML are complementary, not competing

ACROSS THE THREE EXAMPLE DATASETS

Unsupervised detection works in **one of three** regimes

Fraud

TOO EXTREME

0.17% positives, and the label mixes every defect. Best F1 of seventeen detectors is 0.35 (Deep SVDD), best precision 0.34 (HBOS).

Cardio

THE FIT

9.6%, rare and distinct. Isolation Forest F1 0.62, AP 0.70. The regime the detectors are built for.

Water

WRONG REGIME

61% majority target. Every ROC-AUC 0.45 to 0.51: the detectors are at chance. A useful negative result.

The question is less "rules or ML" than "does this dataset have rare, outlying anomalies at all". The synthetic dataset sits in the middle at 17.5% and is where the per-type structure is measured. Pipeline cost narrows the shortlist: One-Class SVM and LOF do not survive the full fraud dataset.

WHAT WAS HARDER THAN THE PLAN ASSUMED

Findings from the build

Each of these became a paragraph in the paper. The recurring lesson: the setup around a detector, the operating point, the regime and the pipeline cost matter as much as the algorithm.

1 The operating point is a lever

Contamination equal to prevalence instead of a fixed 10% moves Isolation Forest fraud precision from 0.01 to 0.20.

2 One regime out of three

Cardio F1 0.62. Fraud, mixed label: nothing usable. Water Quality: at chance.

3 ML is weak on missing values

Imputing removes the signal a not-null rule catches for free: GX recall 1.00 against 0.07 to 0.28 for the classical learned models.

4 GX 1.x is a full API rewrite

Per-row failures need the COMPLETE result format plus an explicit index column; found by probing the installed library.

5 PyOD scores can depend on the batch

ECOD, COPOD and DIF use the scored rows for their statistics, the VAE draws a random code, AE1SVM crashes on a last batch of one row. Recorded adaptations, with 17 batch-independence checks; not upstream-equivalence claims.

6 Drift is not a per-row problem

Once diluted by clean recent rows, the window mean has moved by about a tenth of a standard deviation.

PART 4 OF 4

What the tests **protect**, and what they do not

Verification of the code that produces the numbers, and mutation testing as a way to see its blind spots.

Illustration, not data: a generated still life. It returns on the killed-or-survived slide.

 HISTORICAL RUN, DEFINED SCOPE

Historical verification has a defined scope

Check	Result	Scope
line coverage	1,790 / 1,790	pipeline modules and experiment scripts
branch coverage	350 / 350	same scope
mutation, computing code	95.68%	split, metrics, generator, detectors
mutation, input / output	98.87%	readers, writers, scripts
fault injection	23 / 23	four went through every earlier test
real-data check	pass	three CSV files, manuscript tables match

Survivors (122) count as failures and were classified by reading the changes, not by proof.

These measurements belong to local run **d4304c5**. They cover the core pipeline and experiment scripts, with separate verifier checks.

LIMIT They do not certify the new research modules or establish scientific validity. The run is not independently attested.

1,292

tests, plus 3 on full datasets

97.61%

mutants killed, 4,980 of 5,102 (historical run, core code)

A mutant is a deliberately broken copy of the code

```
# research/alert_budget.py
- flags = scores > info["threshold"]
+ flags = scores >= info["threshold"]
```

A tool makes many copies, each with one small plausible bug, and runs the test suite on every copy. The change here is real: a score **equal** to the training threshold would now raise an alert.

Why bother: coverage says a line ran, not that a test would notice if the line were wrong.

WHAT THE TESTS SEE

score	threshold	original: score > threshold	mutant: score >= threshold	the test expects	test result
0.70	**0.50**	alert	alert	alert	pass
0.30	**0.50**	no alert	no alert	no alert	pass
0.50	**0.50**	no alert	alert	no alert	**FAILS: mutant killed**

The first two tests pass on both versions, so they cannot tell original from mutant. Only the equality case does, and no test covered it before this session. The new test, "a score equal to the training threshold is not an alert", kills this mutant.

The tests are the light. A mutant is killed when a test sees it



A survivor matters because it is either a **behaviour no test protects** (a gap to close with a new test), or a **change that cannot be observed** (an equivalent mutant, next slide). It stays in the count of eligible mutants (the denominator of killed / eligible) until one of the two is shown.

Metaphor, not data: a generated still life. The raised pad casts a shadow; the cracked pad sits flush and casts none.

Some changes cannot be seen, and only the owner may say so

Equivalent mutant

A change that does not alter what the program does, so no test can fail on it. For an integer count `n`, changing `n > 0` to `n >= 1` gives the same answer for every input.

n	n > 0	n >= 1 (mutant)
-1	False	False
0	False	False
1	True	True
2	True	True

Candidates in this work, not approved: a dtype coercion on an array that is already float, zip(strict=) on lists of equal length, flush= on a print.

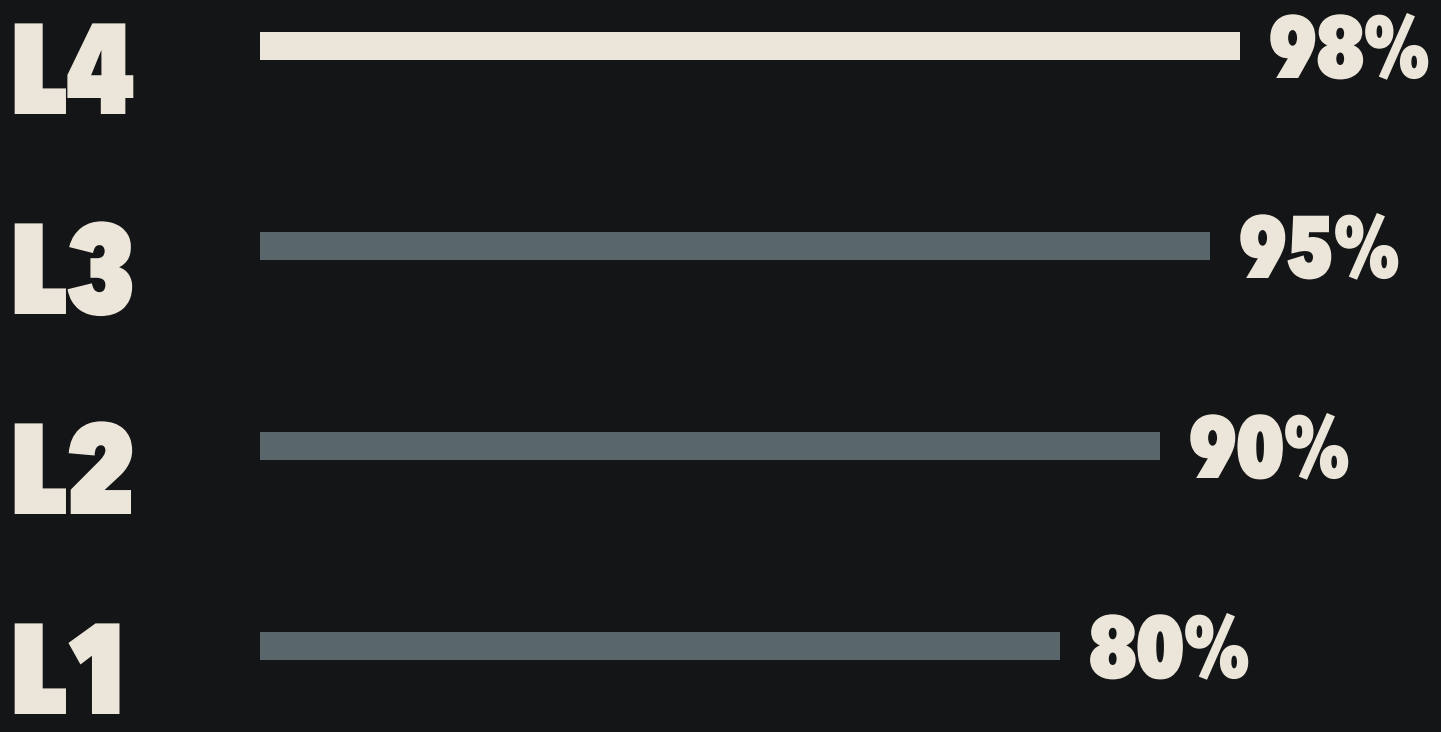
Why approval is required

Calling a survivor equivalent removes it from the denominator and raises the score. If the author of the tests could do that alone, the author would grade their own work. The repository records each equivalence, with its reason, as accepted by the owner.

Status here

The candidates stay in the denominator. The four equivalences accepted earlier for the LSTM autoencoder were accepted by the repository owner, who is also the sole author and reviewer: owner approval, not independent review.

The bar rises with the scientific risk of the code

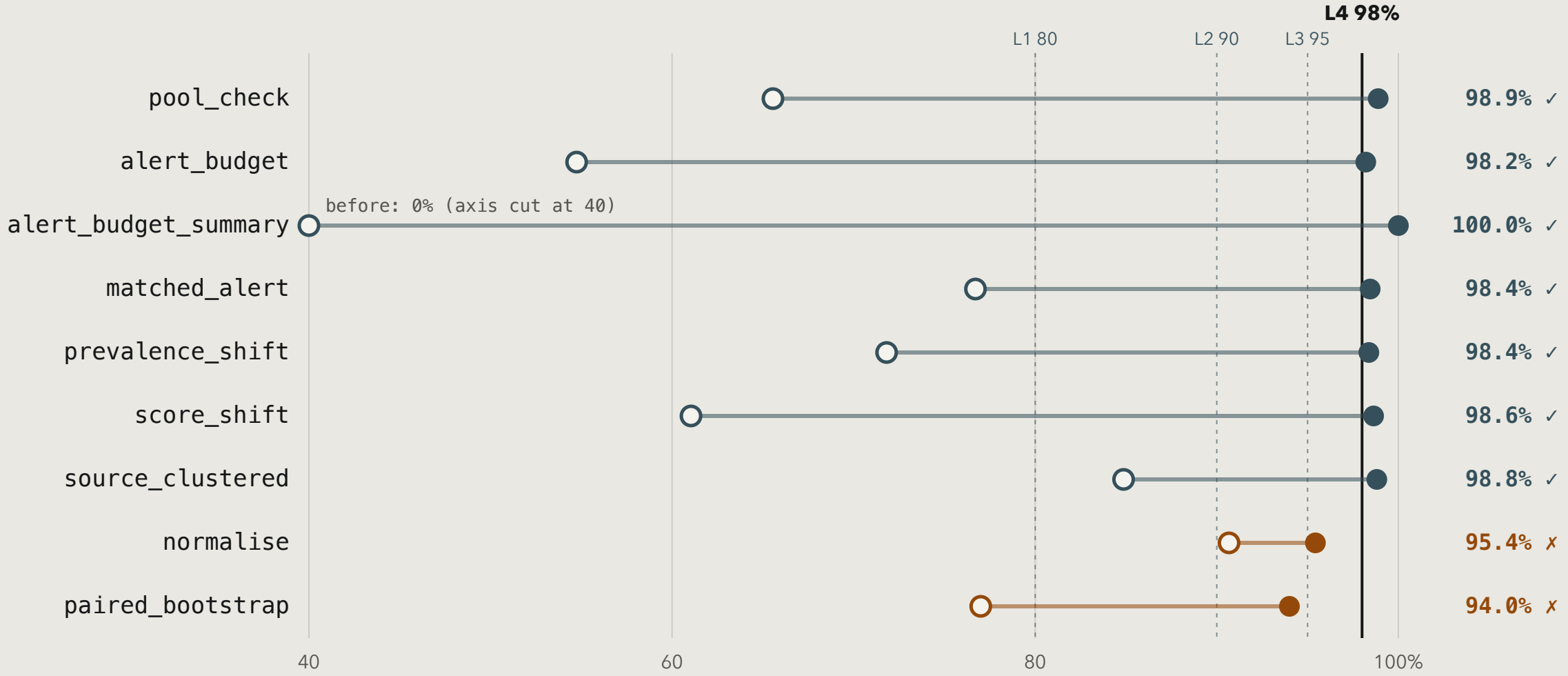


Each module has a risk level, and each level a minimum share of eligible mutants (those counted in the score) that tests must kill.

L4, the highest, is code whose bug would change a reported result. Its bar is **98%**.

The thresholds were not changed and no module was reclassified.

Nine chosen modules: seven reach the 98% bar, two do not



Local run, 2026-11-11. Nine of 82 L4 modules, chosen by the author. Axis from 40%. Not a certification.

One module, two scores: raw 97.61%, adjusted 98.66%

97.61%

raw: 368 of 377 killed

BELOW 98% Every survivor counts.

98.66%

adjusted: 368 of 373

Four mutants removed as equivalent, a smaller denominator by decision. They were accepted by the repository owner, who is the sole author and reviewer: not an independent review.

Five inconclusive mutants stay eligible. No timeout. The adjusted figure depends on a human judgement that raw counts do not need.

WHAT THE EVIDENCE DOES NOT COVER

A focused pass does not certify the repository



26 modules pass

31 modules fail

25 modules incomplete

First complete native campaign, 2026-11-07: 82 L4 modules. 65.5% of L4 mutants killed against 98% required; 7,501 further mutants had no covering test. Bars count modules, the percentage counts mutants, and the 98.66% of the previous slide covers one module of the 82. Not re-measured here.

OPEN

31 L4 modules below threshold and 25 incomplete in that campaign.

OPEN

Survivors that look inert (dtype coercions, zip strict) await an owner decision.

FOUND

Separate from mutation scores: the minimum anomaly count, 3, is below the 6 a stratified 70 / 15 / 15 split needs. No archived result changes.

The release gate fails and stays closed. No full campaign was run in this phase.

WHERE IT STANDS

Experiments run, paper in IMRaD form

DONE

held-out split, 5 seeds, chronological and LSTM pilots, cost study

pool of 152 sets (#66), typed injection on 42 tables (#65)

benchmark on the 100 labelled pool problems

deep rows (#64), multi-schema generator (#67)

IMRaD restructuring (#68, #69, #85); bibliography audit remains #101

decision-level study under shift (#63)

complete registry of 17 detectors, every study rerun

every study extended to the 152-dataset pool, six hypotheses

NEXT: P0 REMEDIATION

Detector and source audits, source-aware statistics, archive replay, publication corrections (issues #89, #91, #100, #103).

The roadmap preserves T001 to T415. **91 of 415** are checked as of 2026-11-11, including T163 to T171 of #94. T172, T062 and T063 remain open, as do the issues.

Optional P2 research is outside the active scope.

TAKEAWAY

Rules and machine learning are complementary **by anomaly type.**

Across the three examples, unsupervised detection pays off only on Cardio, where anomalies are rare outliers. On 100 labelled pool problems, rankings need train-only revalidation. On 42 real tables with injected types, the synthetic pattern holds only in part, which is why the real-data matrix is the differentiator.

The pipeline, the 152-dataset pool, the protocols and the verification layer are in place. The IMRaD write-up and the remaining studies are what remains. The photograph is the cover illustration again: one raised pad among 312.